

Section 7.7: Types of Processing

Data can be processed by a computer in three main ways. These are:

- 1 **Batch Processing**
- 2 **Online Processing**
- 3 **Real-time Processing**

Key Concepts of this section:

- # Understand the **differences between** the three main types of computer processing.
- # Be able to discuss what each type of processing is **typically used for** and give examples.

Data Processing

What is data processing?

- # Data processing is when raw **data is fed into a computer** and then **analysed and changed into useful information**. This useful information can be output and used.
- # Examples of data processing include the following:
 - **Sorting data (ascending / descending)**
 - **Searching / Querying data**
 - **Converting data into another form (e.g. numbers into graphs)**
 - **etc.....**



Key Words:

Processing, Sorting,
Searching, Querying,
Converting



Computers 'process' data

Types of Processing



Key Words:

Batch, Online, Real-time,

As stated above, there are three different types of computer processing:

- **Batch Processing**
- **Online Processing**
- **Real-time Processing.**

1 Batch Processing

This is where a **number of similar jobs or tasks** are **collected together (batched)** and **processed** by a computer **all at once** rather than individually.

For example - producing **electricity bills** for consumers requires **similar calculations**. These calculations can be batched together and carried out all at once.

The collected jobs are loaded into the computer by a human but from there the computer takes over and finishes the processing **without human intervention**.

Batch processing can take a **long time to perform**. This is due to the large number of jobs that are being processed.

Batch processing is **not suitable for tasks** that need to be **processed right away**.

Batch processing is used in the following areas:

- **Producing bills (e.g. electricity, water, gas etc)**
- **Payroll**
- **Stock Control**
- **Producing monthly credit card statements**
- **Processing bank cheques.**
- **Marking multiple choice examination papers**

Examples:



Household bills are Batch Processed



Employee pay-slips are Batch Processed

Advantages / Disadvantages of Batch Processing

Advantages	Disadvantages
Jobs can be processed without a human	Batched jobs can take a long time to

present.

Long jobs can be left processing overnight when everyone has gone home.

process.

'**Batching**' the input data to feed into the system can take a long time.

NOTE!

Examples of batch processing are discussed in greater detail in unit [7.11 - Applications in Finance](#)



Multiple choice exams are Batch Processed when marked

2 Online Processing

- # This type of processing is where a **user communicates directly with the computer system**.
- # When the user makes a **change to the data** stored in the computer, the **system will automatically update and re-process**. This ensures that the system always contains up-to-date information.

For example - Music Concert booking systems need to use online processing. This is because the system needs to **constantly recalculate how many seats are left** as people reserve their tickets.

If the system did not instantly update then people could be paying for seats that are actually no longer available.

- # Online processing is used in the following areas:

- **Booking systems / Ticket reservations** (airlines, cinema, theatre, concerts)
- **ATM's** (recalculates how much money is left in the account)
- **Purchasing goods on the Internet** (e.g. Amazon recalculates how many items are left in stock)

Examples:

Bromma ➔ Malmö	
✈ 29 feb 2007	
Flight:	
Departure	Arrival
▶ 13:15	14:20
15:15	16:20  Full
16:15	17:20

Online processing automatically updates

Advantages / Disadvantages of Online Processing	
Advantages	Disadvantages
Processing is carried out automatically with no need for human supervision.	Needs virus/hacker security (especially for online booking systems).



Online processing allows systems to make sure that concerts are not overbooked

3 Real-time Processing

- # Real-time processing is similar to Online processing in the sense that the computer system automatically updates it's data as changes are made.
- # The difference is that real-time processing often **uses sensors rather than human input** in order to obtain it's data. Real-time processing is used with **control systems**.
- # This type of processing is carried out in **real-time (immediately)**. This means that the computer **responds to inputs without any delays**.

After processing is complete, the system will send instructions to '**control devices**' telling them what to do.

- # Real-time processing is used in the following areas:
 - **Warning systems on aircraft**
 - **Guiding missiles to their target (updates coordinates etc)**
 - **Automatic greenhouses**
 - **Burglar alarms**
 - **Alarm systems in hazardous zones (e.g. monitoring of radiation levels in power stations)**
 - **Flood warning systems.**

Advantages / Disadvantages of Real-Time Processing	
Advantages	Disadvantages
Can carry out jobs that humans are unable to (e.g. monitoring radiation).	Very expensive to buy.

Examples:

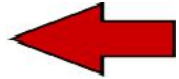


Aircraft warning systems are updated immediately in 'real-time'

NOTE!

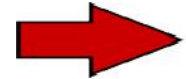
Real-time processing is used in computer control and we covered this in greater detail in unit [7.4 - Control Applications](#).

System failure can cost lives
(e.g. faulty airplane warning system)



**[Previous - Ways in which ICT is used -
Part 2 main menu](#)**

**[Next - Work-Based Communication
Applications](#)**



Links to Theory Units:

[Section 1: Types and Components of Computer Systems](#)

[Section 4: Computer Networks](#)

[Section 7: The Ways in which ICT is used](#)

[Section 2: Input and Output Devices](#)

[Section 5: Data Types](#)

[Section 7: Part 2](#)

[Section 3: Storage Devices and Media](#)

[Section 6: The Effects of Using ICT](#)

[Section 8: Systems Analysis and Design](#)

Links to Practical Units:

[Section 9: Communication](#)

[Section 12: Integration](#)

[Section 15: Web site Authoring](#)

[Section 10: Document Production](#)

[Section 13: Output Data](#)

[Section 16: Presentation Authoring](#)

[Section 11: Data Manipulation](#)

[Section 14: Data Analysis](#)

[Back to top](#)